



Day 38

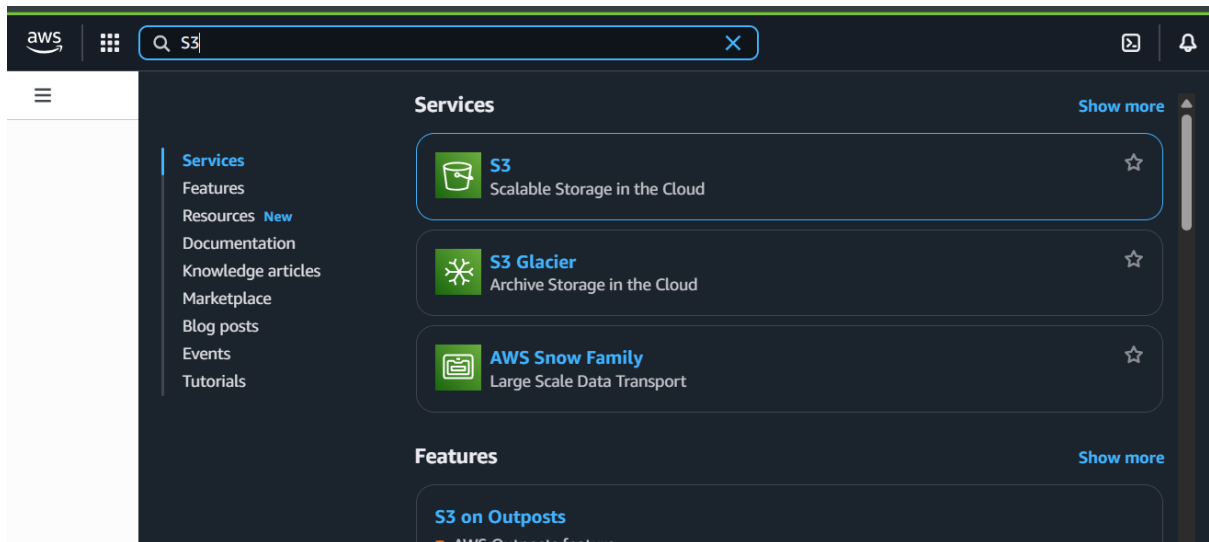


“CLOUD SECURITY”

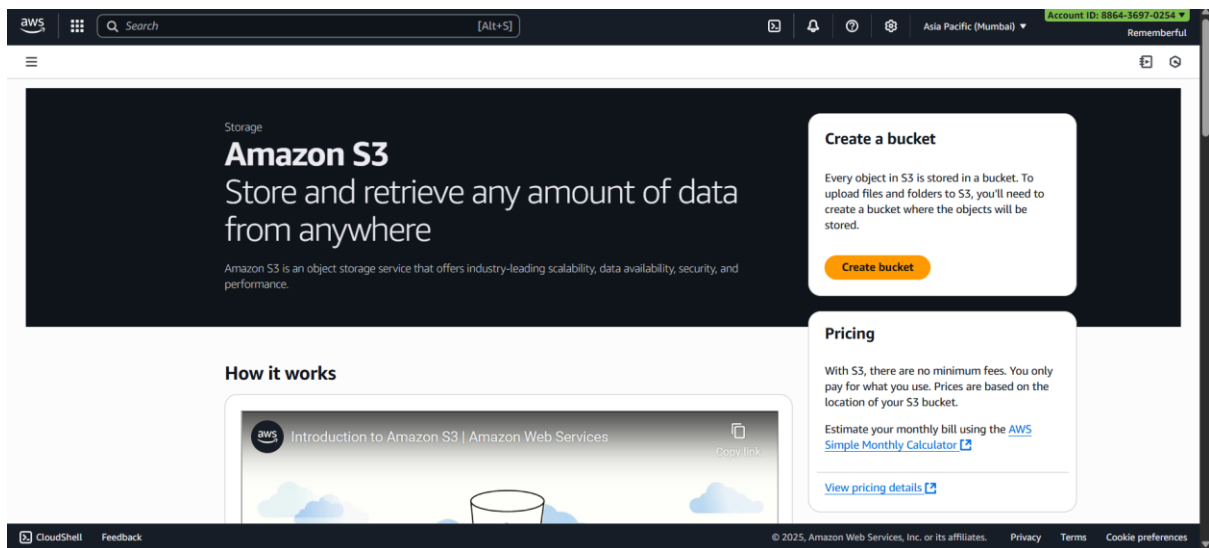
Restricting access to AWS S3 buckets using ACL and Bucket Policy:

Steps:

Look for S3:



Following window will appear:



Click on “Create Bucket”, following form appears:

The screenshot shows the AWS 'Create bucket' form. The top navigation bar includes the AWS logo, a search bar, and account information for 'Asia Pacific (Mumbai)' with account ID '8864-3697-0254'. The breadcrumb trail is 'Amazon S3 > Buckets > Create bucket'. The main heading is 'Create bucket' with a sub-note 'Buckets are containers for data stored in S3.' The 'General configuration' section is active, showing the 'AWS Region' as 'Asia Pacific (Mumbai) ap-south-1'. Under 'Bucket type', 'General purpose' is selected with a description: 'Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.' The 'Directory' option is unselected. The 'Bucket name' field contains 'myawsbucket' with a note: 'Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). Learn More'. Below this, 'Copy settings from existing bucket - optional' is shown with a 'Choose bucket' button and the format 's3://bucket/prefix'. The 'Object Ownership' section is partially visible at the bottom.

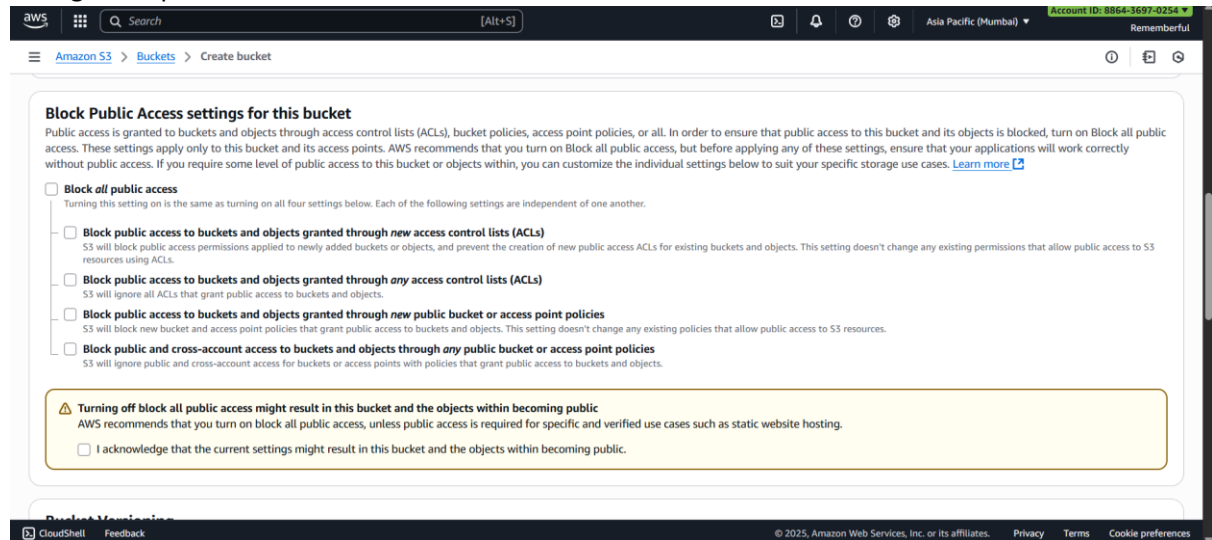
Fill it:

This screenshot shows the same 'Create bucket' form, but the 'Bucket name' field is now filled with 'bkbik'. The 'General purpose' bucket type remains selected. The 'Copy settings from existing bucket - optional' section is still present with the 'Choose bucket' button. The 'Object Ownership' section is more visible, showing the 'ACLs disabled (recommended)' option selected, with a description: 'All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.' The 'ACLs enabled' option is unselected. A warning message states: 'We recommend disabling ACLs, unless you need to control access for each object individually or to have the object writer own the data they upload. Using a bucket policy instead of ACLs to share data with users outside of your account simplifies permissions management and auditing.'

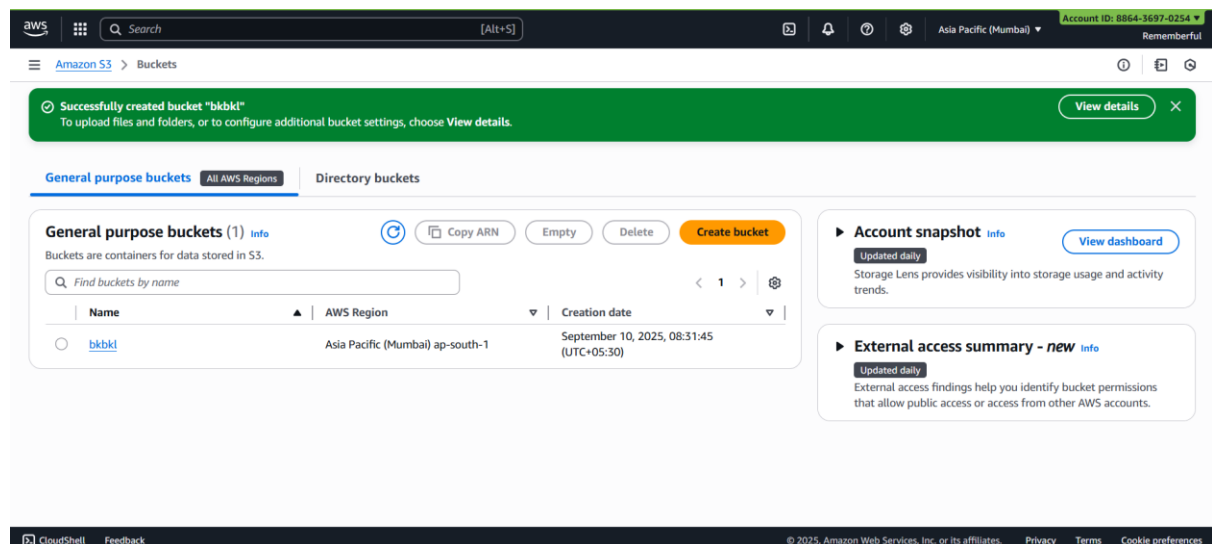
Change it as “ACLs enabled”:

This screenshot shows the 'Create bucket' form with 'ACLs enabled' selected under 'Object Ownership'. The description for 'ACLs enabled' is: 'Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.' The 'Bucket owner preferred' option is also selected under 'Object Ownership', with a description: 'If new objects written to this bucket specify the bucket-owner-full-control canned ACL, they are owned by the bucket owner. Otherwise, they are owned by the object writer.' The 'Object writer' option is unselected. A note at the bottom states: 'If you want to enforce object ownership for new objects only, your bucket policy must specify that the bucket-owner-full-control canned ACL is required for object uploads. Learn more'. The 'Block Public Access settings for this bucket' section is partially visible at the bottom.

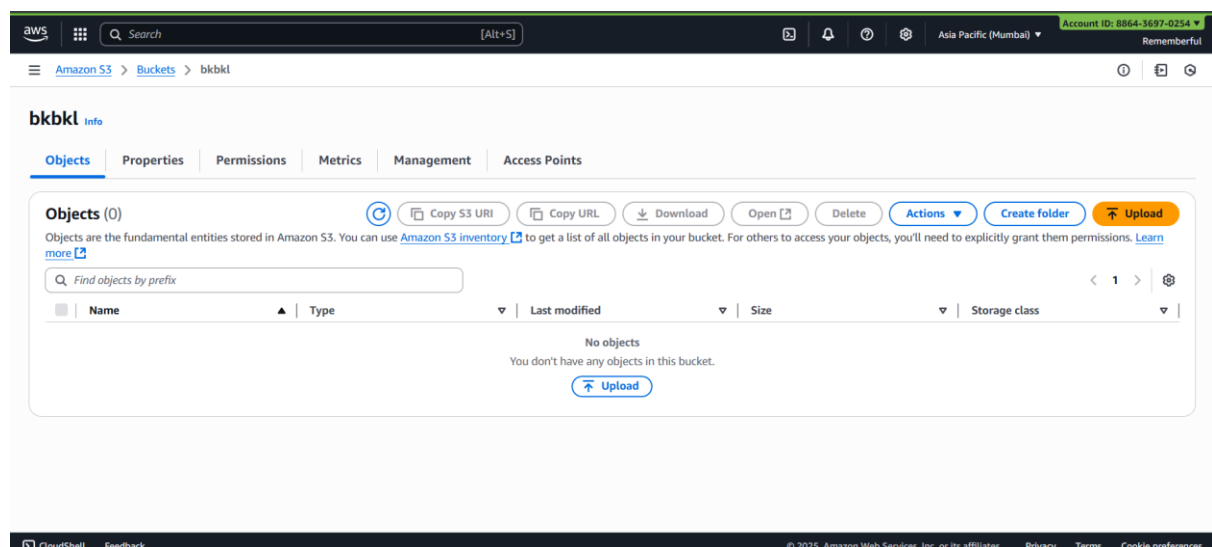
Change it to public:



Following confirmation will occur:



Open the bucket:



Go to “permissions” section: following section appears

The screenshot shows the AWS Management Console interface for an Amazon S3 bucket named 'bkbkl'. The 'Permissions' tab is selected, displaying the 'Permissions overview' section. This section includes an 'Access finding' link, a 'Block public access (bucket settings)' section with an 'Edit' button, and a 'Bucket policy' section with 'Edit' and 'Delete' buttons. The 'Block public access' section shows a warning icon and the text 'Off', with a link to 'Individual Block Public Access settings for this bucket'.

Scroll down to “Access control list (ACL)”

The screenshot shows the 'Access control list (ACL)' section for the 'bkbkl' bucket. It includes a warning message: 'The console displays combined access grants for duplicate grantees. To see the full list of ACLs, use the Amazon S3 REST API, AWS CLI, or AWS SDKs.' Below this is a table with three columns: 'Grantee', 'Objects', and 'Bucket ACL'. The table lists four grantees: 'Bucket owner (your AWS account)', 'Everyone (public access)', 'Authenticated users group (anyone with an AWS account)', and 'S3 log delivery group'. Each grantee has associated permissions for 'Objects' and 'Bucket ACL'. At the bottom, there is a 'Cross-origin resource sharing (CORS)' section with an 'Edit' button.

Grantee	Objects	Bucket ACL
Bucket owner (your AWS account) Canonical ID: 7b1be54aeb3c258dcdfcb5a441ee344179960318247860ad75b598226eafaa40	List, Write	Read, Write
Everyone (public access) Group: http://acs.amazonaws.com/groups/global/AllUsers	-	-
Authenticated users group (anyone with an AWS account) Group: http://acs.amazonaws.com/groups/global/AuthenticatedUsers	-	-
S3 log delivery group Group: http://acs.amazonaws.com/groups/s3/LogDelivery	-	-

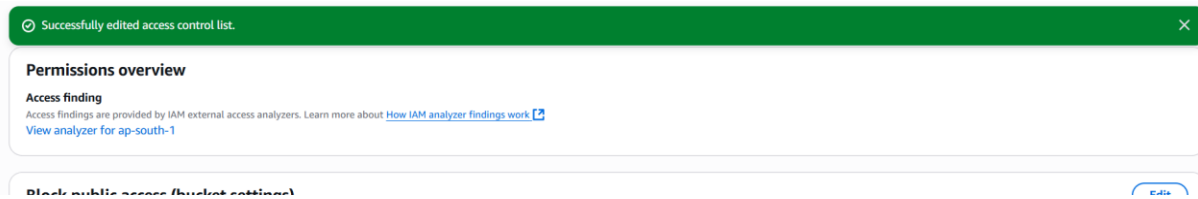
Click on “edit” button: following options will occur.

The screenshot shows the 'Edit access control list (ACL)' page for the 'bkbkl' bucket. It displays the same table as the previous screenshot, but with checkboxes for 'List' and 'Write' permissions for each grantee. The 'Bucket owner' has both 'List' and 'Write' permissions checked. The 'Everyone (public access)', 'Authenticated users group', and 'S3 log delivery group' have both 'List' and 'Write' permissions unchecked. There is an 'Add grantee' button at the bottom.

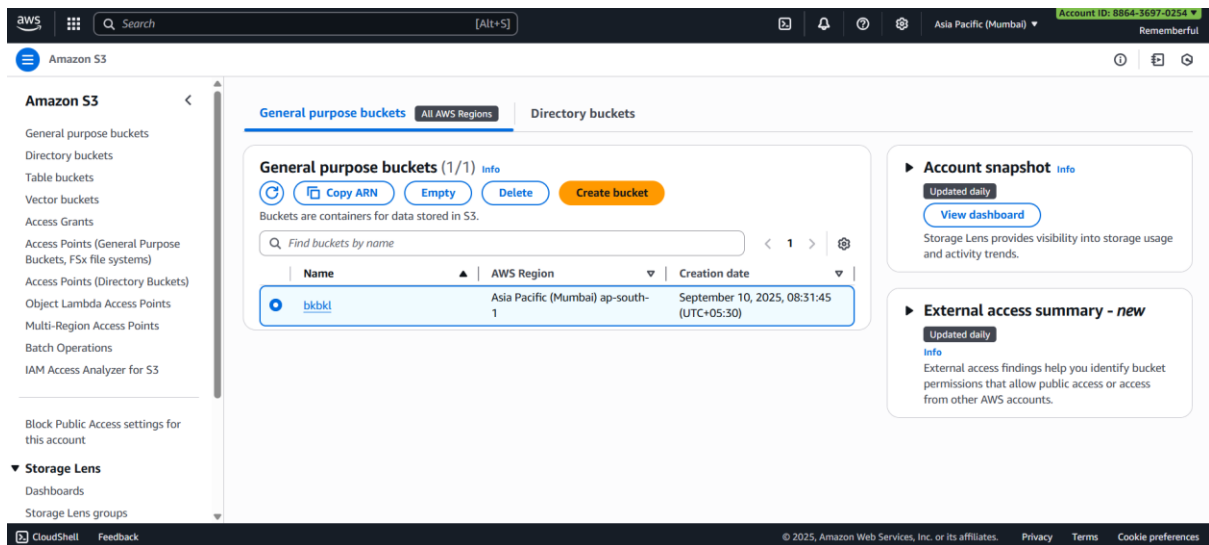
Grantee	Objects	Bucket ACL
Bucket owner (your AWS account) Canonical ID: 7b1be54aeb3c258dcdfcb5a441ee344179960318247860ad75b598226eafaa40	☒ List ☒ Write	☒ Read ☒ Write
Everyone (public access) Group: http://acs.amazonaws.com/groups/global/AllUsers	☐ List ☐ Write	☐ Read ☐ Write
Authenticated users group (anyone with an AWS account) Group: http://acs.amazonaws.com/groups/global/AuthenticatedUsers	☐ List ☐ Write	☐ Read ☐ Write
S3 log delivery group Group: http://acs.amazonaws.com/groups/s3/LogDelivery	☐ List ☐ Write	☐ Read ☐ Write

Keep it as it is, all List, Write, Read, write section as default.

Save, it following confirmation will occur:



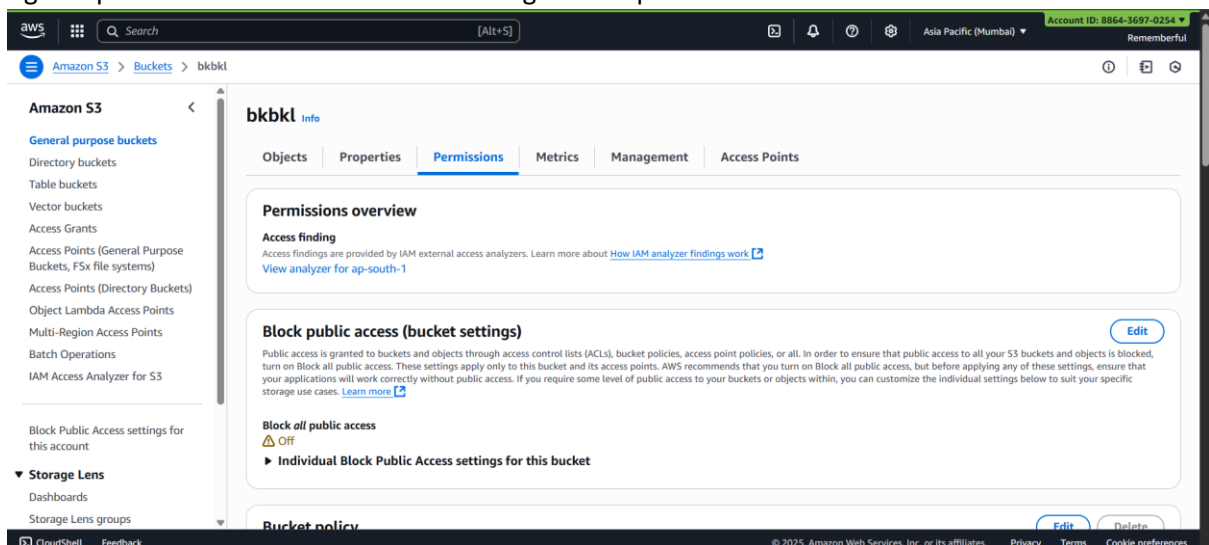
Go to the bucket section and select the bucket and ARN:



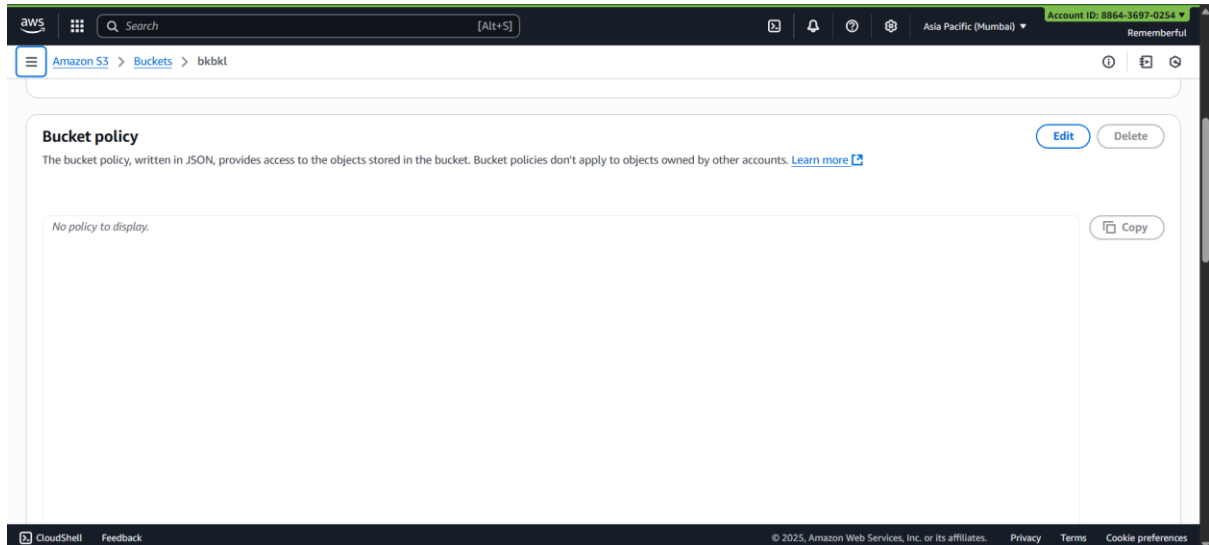
Copy paste the ARN in the notepad:

```
arn:aws:s3:::bkbkl
```

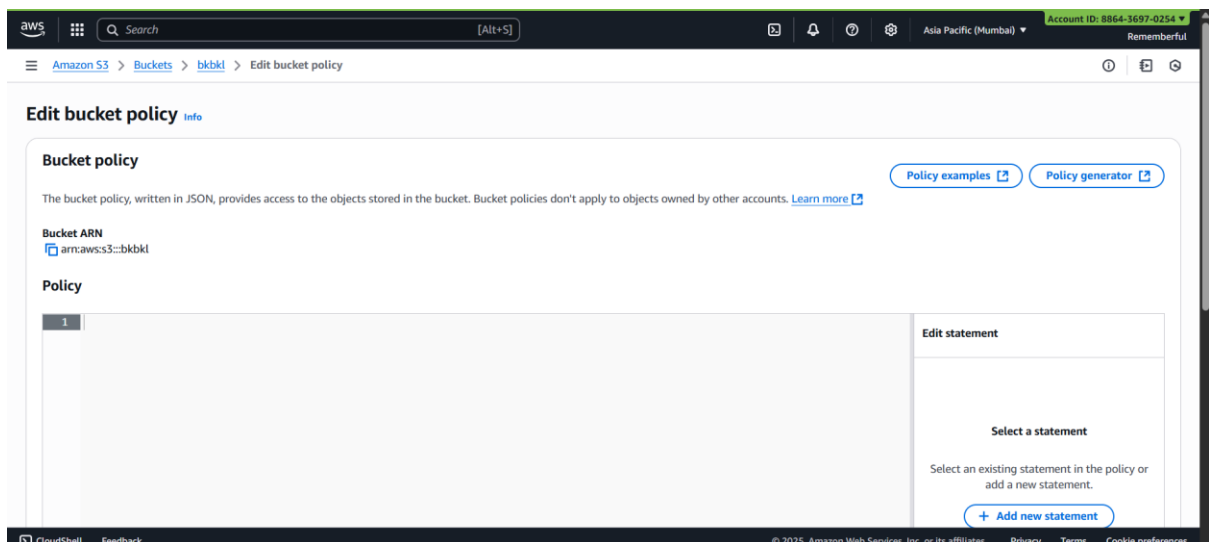
Again open the bucket we had created and go to the permissions section:



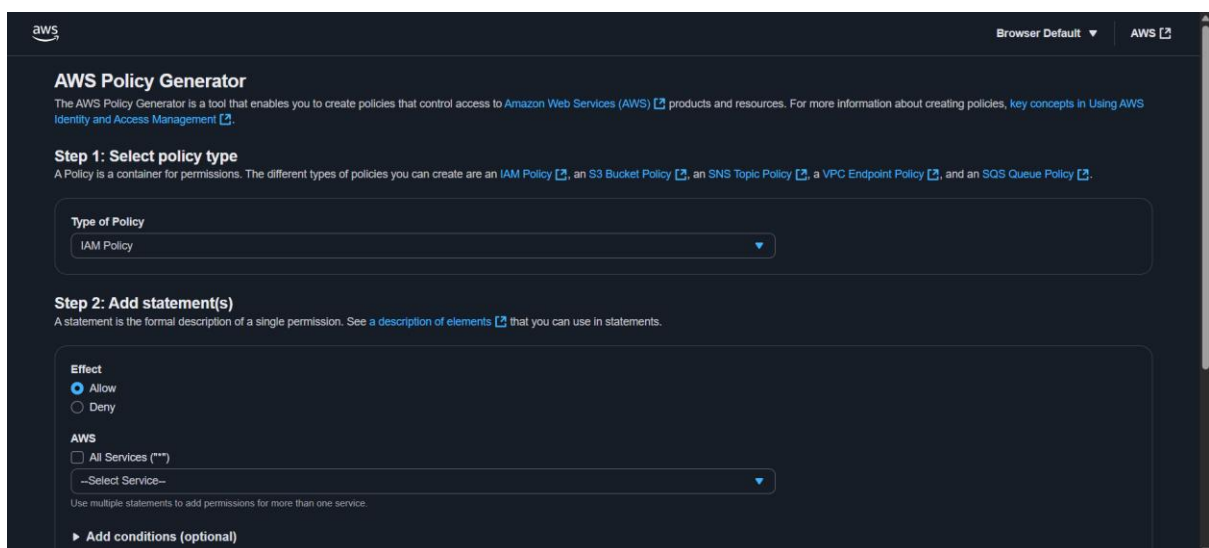
Scroll down and reach “Bucket Policy”



Click on “edit” button: following window will appear



Click on “Policy generator” button: following screen will appear.



Make changes as shown:

The screenshot shows the 'Add statement(s)' step in the AWS IAM console. At the top, a dropdown menu is set to 'S3 Bucket Policy'. Below this, the 'Step 2: Add statement(s)' section is active. It includes a description: 'A statement is the formal description of a single permission. See a [description of elements](#) that you can use in statements.' The form contains several fields: 'Effect' with 'Allow' selected; 'Principal' with a text input containing '*'; 'Actions' with 'All Actions (*)' selected; and 'Amazon Resource Name (ARN)' with a text input containing 'arn:aws:s3:::bkbkl'. There is also an 'Add conditions (optional)' link and an 'Add Statement' button at the bottom.

Following pop up comes:

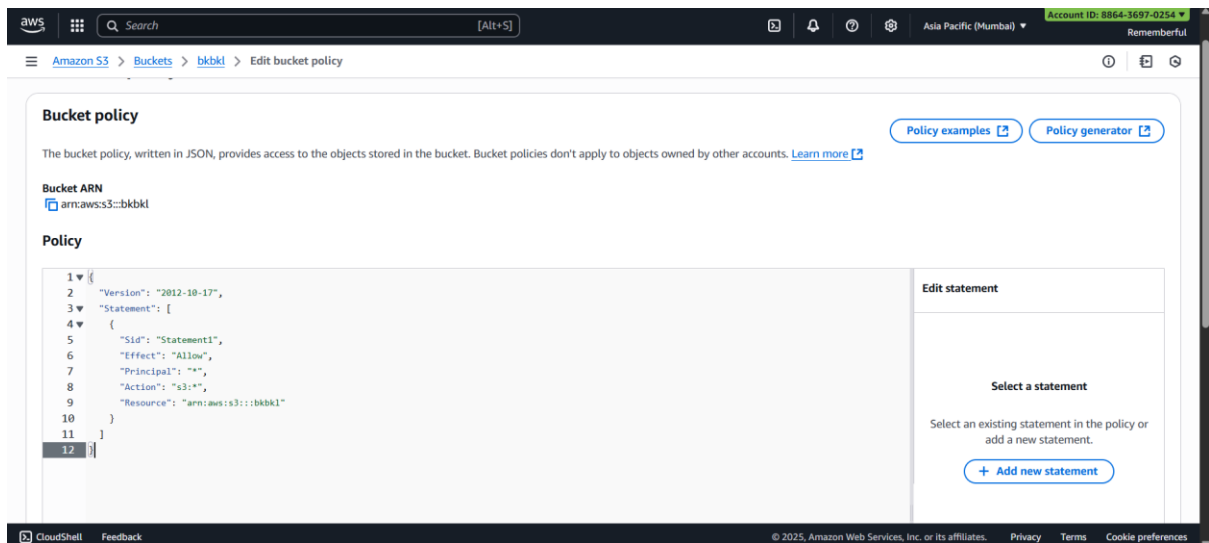
The screenshot shows a 'Policy JSON Document' pop-up window. It has a close button in the top right corner. The main text says: 'Click below to edit. To save the policy, copy the text below to a text editor. Changes made below will **not be reflected in the policy generator tool**.' Below this is a code editor with a line number gutter on the left (lines 1-12) and a copy icon on the right. The JSON content is:

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Sid": "Statement1",
6       "Effect": "Allow",
7       "Principal": "*",
8       "Action": "s3:*",
9       "Resource": "arn:aws:s3:::bkbkl"
10    }
11  ]
12 }
```

 At the bottom right of the code editor, it says '1:1 JSON'. Below the code editor is a disclaimer: 'This AWS Policy Generator is provided for informational purposes only, you are still responsible for your use of Amazon Web Services technologies and ensuring that your use is in compliance with all applicable terms and conditions. This AWS Policy Generator is provided as is without warranty of any kind, whether express, implied, or statutory. This AWS Policy Generator does not modify the applicable terms and conditions governing your use of Amazon Web Services technologies.' At the bottom of the pop-up are two buttons: 'Close' and 'Copy Policy'.

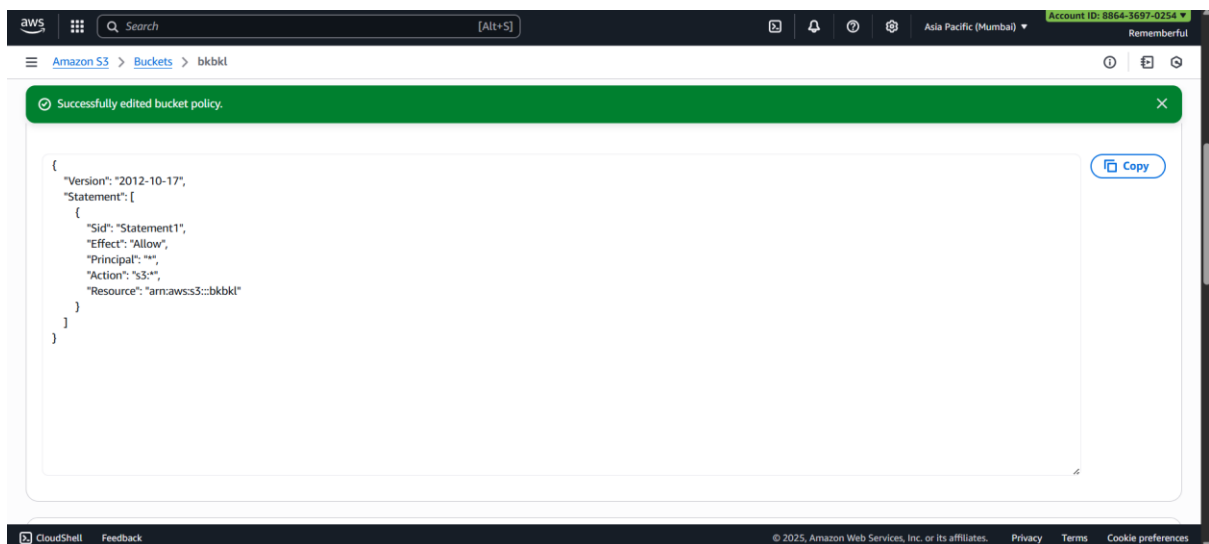
Copy the policy.

Come back to the earlier page:



Click on save changes.

Following confirmation will occur:



--The End--